

CivicFlow PRD – Indian Government Civic & Services Super-App

1. Overview

CivicFlow is a proposed Indian government-facing Android + web application that unifies civic issue reporting and access to key central/state e-governance services (Aadhaar, PAN, Passport, and others) into a single, location-aware platform, similar in spirit to UMANG but with a stronger focus on city-level issue resolution and citizen trust.^{[1][2][^3]}

The app is intended for deployment on an India-hosted cloud (or government cloud) and includes a complete backend (APIs, databases, integrations, analytics) along with citizens' mobile/web frontends and administrative portals for government staff.

2. Problem, Vision and Objectives

2.1 Problem statement

Citizens currently navigate multiple portals and apps to access government services for Aadhaar, PAN, Passport, EPFO, railways, etc., which increases friction and excludes less tech-savvy users.^{[2][1]}

At the same time, civic complaint apps like Swachhata-MoHUA and BBMP Fix My Street focus narrowly on sanitation or potholes and often suffer from UX issues (e.g., confusing maps, mandatory photo flows that sometimes fail) and limited transparency around resolution.^{[4][5][^6]}

There is no single cohesive experience that combines everyday civic issue reporting, key government services, and citizen engagement in a location-smart, mobile-first interface.

2.2 Vision

Provide a single, trustworthy, location-aware app where any Indian resident can:

- Report a civic issue in under 30–60 seconds, with automatic routing to the responsible ward/department and transparent tracking until closure.^{[7][8][^4]}
- Access a curated set of high-impact central/state government services (Aadhaar, PAN, Passport, DigiLocker documents, a limited set of state/ULB services) without needing to install or remember multiple apps.^{[1][2]}
- Receive proactive communication from government (MyGov-style engagement, campaigns, surveys and feedback) to strengthen participatory governance.^{[9][10][^11]}

2.3 Objectives

- Reduce friction in reporting civic issues and accessing priority services compared with existing single-purpose apps.^{[12][4]}
- Improve resolution rates and citizen satisfaction through smart routing, SLA tracking, and evidence-backed closure for civic complaints.^{[4][7]}
- Provide a modular backend so new services or departments (e.g., state e-district, municipal water board) can be onboarded with minimal development.

3. Competitive and Reference Landscape

3.1 UMANG

UMANG is a Government of India super-app that aggregates 100+ services (central, state, local), including Aadhaar, PAN, Passport, DigiLocker documents, EPFO, and many others, across mobile and web.^{[3][2][^1]}

Its key strengths are breadth of coverage, single sign-on across many schemes, and deep integration with central platforms like DigiLocker, but its IA and UX can be overwhelming for first-time users due to the sheer number of services.

3.2 Swachhata-MoHUA

Swachhata-MoHUA is MoHUA's official civic issue app for sanitation-related problems, allowing users to post issues like garbage dumps which are routed to city corporations and assigned to ward-level sanitary inspectors.^{[6][13][^12]}

It demonstrates an end-to-end issue routing and closure model but is scoped largely to cleanliness/sanitation and does not cover broader civic issues or generic government services.

3.3 BBMP Fix My Street

BBMP Fix My Street is Bengaluru's official app for reporting potholes and other civic amenity issues, with geo-tagged photo complaints, SMS updates, registration-optional reporting, and dashboards for officers who can transfer and close complaints with photos.^{[5][4]}

User reviews highlight UX issues around photo capture and awkward world-map based location selection, illustrating the need for more forgiving, Indian-city-specific location UX.^[^5]

3.4 SeeClickFix and other 311 platforms

SeeClickFix in North America offers “See a problem, Click the map, Fix the issue” flows with smart routing from issue type and location into municipal 311 systems, real-time status, and “watch areas” for following local issues.^{[8][7]}

These platforms demonstrate the value of combining location, transparency, and notifications to build trust between residents and local governments.

3.5 MyGov

MyGov is India's national citizen engagement platform for consultations, polls, quizzes, and campaigns, with tens of millions of users participating in policy feedback and governance-related activities.^{[10][11][^9]}

It is a strong reference for engagement, but does not directly handle service delivery or issue-level resolution.

3.6 Differentiation for CivicFlow

CivicFlow aims to:

- Combine UMANG-style multi-service access with Swachhata/BBMP-style local issue resolution and MyGov-style engagement in a single app.^{[2][4][^1]}
- Prioritize a simpler IA by focusing on a curated set of high-impact services rather than all government services in V1, with clear verticals (Identity, Finance & Work, Travel & Transport, Civic Issues, Local Services).
- Offer robust, ward-aware location UX and a modern Android experience that addresses usability pain points seen in existing civic apps.^{[4][5]}

4. Scope and Non-Goals

4.1 In-scope (V1-V2)

- Citizen-facing Android app and responsive web frontend.
- Location-aware civic issue reporting across a configurable taxonomy (roads, waste, streetlights, drainage, water, encroachment, public safety, etc.).^{[12][4]}
- Access to a scalable catalog of Indian government services (aiming to cover all eligible central, state and ULB services over time) through:
 - Official APIs where they exist and third-party integration is permitted.
 - Deep links or webviews into existing official portals and UMANG flows where APIs are not publicly available.^{[1][2]}
- Admin/operations portals for municipal/ULB staff and selected state/national back-office teams.
- Notification and communication layer (push/SMS/email) for status updates and engagement.
- Multi-language UI (starting with English + Hindi, extensible to regional languages) and accessibility support.

4.2 Explicit non-goals (V1)

- Replacing UMANG or existing official apps; CivicFlow should integrate with and route into them where possible.^{[2][1]}
- Storing or replicating sensitive identity data like Aadhaar numbers beyond what is minimally required for specific service requests, and never storing biometrics.
- Direct access to UIDAI, PAN, or Passport master data unless formally onboarded with appropriate legal and security clearances; until then, only officially documented APIs or deep links are used.^{[1][2]}
- Handling emergency response (police, fire, ambulance); these remain under existing 112/100/other emergency systems.

5. User Personas and Key Use Cases

5.1 Personas

- **Citizen User (Urban/Rural)** – Any resident with a smartphone needing to:
 - Report civic issues near their home or workplace.
 - Access Aadhaar/PAN/Passport and other services without juggling multiple apps.^{[12][1]}
- **Ward/ULB Officer** – Municipal official responsible for triaging, assigning, and closing issues in a ward or department (roads, sanitation, water).^{[5][4]}
- **Department Supervisor/Commissioner** – Monitors KPIs, ensures SLA compliance, reallocates resources across wards.
- **State/National Back-Office User** – Manages configuration (service catalog, deep links, routing) and monitors adoption and feedback.

5.2 Key citizen use cases

- Report a pothole with photo and GPS in under a minute and track its status through “Received → In Progress → Resolved” with proof of repair.^{[4][5]}
- Raise a garbage complaint similar to Swachhata-MoHUA, but with consistent UX for all civic categories.^{[6][12]}
- Check PAN application status, link PAN with Aadhaar, or download e-PAN through UMANG-compatible flows.^[^1]
- Check Aadhaar services (update appointment info, virtual ID, download e-Aadhaar) via UMANG or UIDAI deep links.^[^2]
- Check Passport application status or schedule an appointment, routed into the official Passport Seva or UMANG interface.^{[2][1]}
- Participate in MyGov-style surveys or campaigns within CivicFlow for local and national initiatives.^{[11][9][^10]}

5.3 Key staff use cases

- Ward engineer views a map/list of all open issues in their ward, filtered by category and SLA breach risk, and updates field work with geo-tagged photos.^[5]^[4]
- Sanitation inspector sees issues forwarded from Swachhata-type categories in CivicFlow and updates status while on the field.^[6]^[12]
- Commissioner reviews dashboards of daily complaint volume, resolution rates, and hotspots across wards and departments, similar to BBMP dashboards.^[^4]

6. User Journeys and Flows

6.1 Civic issue reporting journey

1. Launch app → Home

- Detect user location (with consent) and show the city/ULB and nearest ward boundary on the map.

2. Tap “Report an issue”

- Choose a category (Road, Garbage, Streetlight, Water, etc.); show contextual icons.

3. Capture evidence

- Take a photo or upload from gallery; app compresses and geo-tags the media.

4. Confirm location

- Auto-place a pin using GPS; user can drag to correct; map defaults to current city rather than world view to avoid BBMP UX criticism.^[^5]

5. Describe issue

- Short description, optional landmark and contact details.

6. Submit

- Show ticket ID and estimated response time; optionally allow anonymous or logged-in submissions.

7. Track

- Timeline view with status changes, comments from staff, and closure photos.

6.2 Government service access journey (example: PAN)

1. Launch app → Services tab

- Service catalog grouped by vertical: Identity, Finance & Work, Travel, Civic & Local, etc.

2. Select PAN services

- Options like “Apply for new PAN,” “Check PAN status,” “Download e-PAN,” etc., modeled on UMANG PAN services.^[^1]

3. Perform action

- If integrated API exists: perform within CivicFlow.
- Else: open secure webview or deep link into UMANG or the official portal with pre-filled context where supported.^{[2][1]}

4. Receive confirmation

- Show reference ID/status and optionally store it in the user's "My Requests."

6.3 Engagement journey

- Citizens can access MyGov-style polls, discussions, and campaigns surfaces within CivicFlow's "Engage" tab or as banners, routing to MyGov or embedding its UX where integration is available.^{[9][10][^11]}

7. Functional Requirements

7.1 Authentication and identity

- Mobile-number based sign-in with OTP (primary).
- Optional account linking to DigiLocker/UMANG/ID tokens where official integrator status is granted.^{[1][2]}
- Support for guest reporting of civic issues (with some limitations: no history sync across devices).
- Profile management: name, preferred language, city, ward/home address, preferred channels.

7.2 Civic issue module

- Configurable category taxonomy per city/state (e.g., pothole, garbage, streetlight, sewage, water supply, encroachment, tree fall, stray animals, etc.).^{[12][4]}
- Multi-media support: photo (mandatory by default but overridable by configuration), video (optional) with compression and offline upload queue.
- Location handling:
 - GPS auto-capture with accuracy display.
 - Manual pin adjustment with city-level bounding box to avoid global map confusion seen in BBMP app.^[^5]
 - Reverse-geocoding to get address + ward/zone.
- Ticket lifecycle: Draft → Submitted → Acknowledged → Assigned → In Progress → On Hold → Resolved → Reopened.
- Duplicate detection: server-side matching based on location radius, category, and time window; suggest joining an existing ticket instead of creating a fresh one.
- Commenting: citizens and officials can exchange messages on a ticket, with moderation controls.
- Attachments: closure photos from staff, auto-tagged with time and location.

7.3 Government service hub module

- Services catalog with meta-data:
 - Name, description, owning ministry/body, tags, category, integration type.
 - Integration types:
 - Direct API integration (rare in V1, where officially supported).
 - Deep link to UMANG service.^[2]^[1]
 - Deep link to official portal (e.g., Passport Seva, PAN portal).
 - Webview wrapper for services that are mobile-web optimized.
- For high-value services (V1 priorities):
 - PAN: apply, status, corrections, e-PAN (UMANG-style set).^[^1]
 - Aadhaar: e-Aadhaar/DigiLocker document retrieval, appointment/center information, virtual ID, limited updates if API permitted.^[^2]
 - Passport: status tracking and appointment links.^[^1]
- Requests and status tracking consolidation in “My Requests,” even when handled by external portals, by allowing manual reference ID entry.

7.4 Admin and operations portals

- ULB/ward staff console:
 - Map and list view of issues, with filters for category, status, SLA, date range, and ward.
 - Bulk actions (assign, reassign, close) with reason codes.
 - Field-staff mobile view for updating issues with comments and photos.
- City/state administrator console:
 - Configuration for categories, SLAs, routing rules, wards/department structure.
 - Role-based access control for officials.
 - Analytics dashboards: volume trends, heatmaps, resolution times, SLA compliance, department performance.^[14]^[4]

7.5 Notifications and communication

- Push notifications for ticket lifecycle events (submission, assignment, comment, status changes, closure), service request updates (where APIs/UMANG allow), and engagement campaigns.^[7]^[2]
- SMS fallback for key events in low-data environments, modeled on BBMP’s SMS updates.^[4]^[5]
- In-app inbox with history of notifications.

7.6 Location and GIS

- Ward/zone boundaries stored in geospatial format (PostGIS), enabling automatic assignment to appropriate ward office based on ticket location.[⁴]
- Heatmaps and cluster views of complaints for admin dashboards, similar to smart-city 311 platforms.[¹⁴]
- “Watch area” subscription like SeeClickFix, where citizens can follow issues within a radius around a chosen location.[⁸][⁷]

7.7 Localization and accessibility

- Multi-language UI with at least English and Hindi at launch; extensible to major regional languages via a translation framework.[²]
- WCAG-aligned color contrast, large touch targets, support for screen readers.
- Offline-friendly behaviors: caching of categories, draft ticket storage until connectivity is restored.

8. Non-Functional Requirements

- **Scalability:** Able to handle peak loads for large metros (e.g., concurrent use across millions of residents during campaigns or crises).[²]
- **Availability:** Target 99.5% uptime for core APIs; graceful degradation when specific integrations are down.
- **Security & privacy:**
 - Follow Indian data protection norms and sectoral guidelines; minimize storage of PII beyond what is strictly needed.
 - Encrypt data in transit (TLS 1.2+) and at rest; apply strong access control for admin consoles.
 - Do not store Aadhaar numbers or other sensitive IDs unless mandated and with full compliance; where possible, use DigiLocker tokens or UMANG identity flows instead of raw identifiers.[¹][²]
- **Compliance:**
 - Align with MeitY e-Gov standards and UX4G components where applicable.[¹¹]
 - For hosting on government infrastructure, follow NIC/MeghRaj security baselines.
- **Observability:** Centralized logging, metrics, and alerting; audit trails for all staff actions on civic issues.

9. MVP Definition

9.1 MVP goals

- Launch CivicFlow in 1-2 pilot cities/ULBs with a working civic issue pipeline and a minimal yet valuable set of central services accessible via integrations.
- Prove that citizens can efficiently report and track issues, and that departments can manage and resolve them with measurable improvements.

9.2 MVP feature set

Citizen app/web (MVP)

- Onboarding, mobile-number OTP login, guest mode.
- Home screen with “Report an issue,” “Track my issues,” and “Services” entry points.
- Civic issue reporting flow (photo, category, description, GPS + pin, submit) with basic duplicate detection and timeline tracking.^{[5][4]}
- Services hub with at least:
 - PAN services via UMANG or official deep link (apply, status, e-PAN).^[^1]
 - Aadhaar and Passport status/links via UMANG/official portals.^{[2][1]}
- Notifications for issue lifecycle and essential service updates.

Admin/ULB portal (MVP)

- Ward-level list of complaints, assign/reassign/close.
- Ward definitions uploaded as shapefiles or equivalent.
- Basic analytics: complaint volume, open vs. resolved, SLA breach count.

Backend (MVP)

- REST/GraphQL APIs for tickets, users, services, and notifications.
- Database with geospatial support.
- Integration layer for sending users into UMANG and official portals.
- FCM/SMS integration.

9.3 Out-of-scope for MVP (future releases)

- AI-based classification of issues via image/text, voice bots, WhatsApp chatbots, and advanced predictive analytics.^{[15][14]}
- Full replication of UMANG’s service catalog; instead, add services incrementally with demand and partnership.^{[3][2]}
- Deep integration with MyGov campaigns beyond simple linking/embedding.

10. Backend Architecture (High-Level)

10.1 Suggested stack

- **Backend framework:** Node.js (NestJS) or Java/Spring Boot REST services, or Python/FastAPI, chosen per implementing agency capabilities.
- **Database:** PostgreSQL with PostGIS extensions for geospatial operations (ward mapping, proximity search).
- **Cache/queue:** Redis for caching and background job coordination; optional message broker (e.g., Kafka/RabbitMQ) for high-volume event pipelines and notification fan-out.
- **Object storage:** S3-compatible storage (e.g., AWS S3, Azure Blob, or NIC cloud equivalent) for complaint media.
- **Auth & identity:** OAuth2/OpenID Connect for staff login (integrate with government IdPs where available); token-based auth for citizen APIs; optional integration with UMANG/India Stack identity components.^{[1][2]}
- **Integration layer:** Micro-service or module that handles outbound calls/webviews into UMANG and other portals, encapsulating differences between services.^{[1][2]}

10.2 Logical components

- **API Gateway:** Single entry point for mobile/web clients; rate limiting and authentication.
- **Civic Service:** Manages complaints, categories, SLA logic, routing rules.
- **Location Service:** Handles geocoding/reverse-geocoding, ward mapping, geo-queries.
- **Government Services Catalog Service:** Stores service meta-data and integration configuration.
- **Notification Service:** Sends push/SMS/email via pluggable providers.
- **Admin/Config Service:** Manages tenants (cities/states), roles, and configurations.

11. Cloud Deployment Model

11.1 Hosting environments

- **Government cloud (preferred for production):** NIC MeghRaj or equivalent Indian government cloud, adhering to MeitY guidelines.^{[1][2]}
- **Commercial cloud (for pilots/prototyping with approvals):**
 - AWS (ap-south-1 Mumbai region) or Azure India regions, to keep data resident in India.

11.2 Deployment topology

- Containerized microservices or modular monolith deployed on Kubernetes (EKS/AKS or NIC Kubernetes service) or a managed PaaS (e.g., AWS Elastic Beanstalk) for simpler setups.
- Managed PostgreSQL (RDS/Aurora, Azure Database for PostgreSQL, or NIC DBaaS) with PostGIS.
- CDN-backed static hosting for web frontend (e.g., CloudFront, Azure CDN, or NIC CDN) with HTTPS termination.
- Separate environments for dev/test/staging/production with CI/CD pipelines and automated tests.

11.3 Monitoring and operations

- Centralized logging (e.g., ELK, OpenSearch, or government-approved equivalent).
- Metrics and tracing (Prometheus/Grafana, OpenTelemetry).
- Regular backups, DR plan with defined RPO/RTO.

12. Risks and Mitigations

- **Integration approvals:** Obtaining official permission to integrate with Aadhaar/PAN/Passport services may be time-consuming; the design therefore treats deep links and webviews into UMANG/official portals as the baseline integration model for early phases.^{[1][2]}
- **Data sensitivity:** Identity and location data are sensitive; CivicFlow minimizes what it stores, encrypts it, and relies on existing national platforms for heavy-duty identity functions.
- **Adoption and overlap:** UMANG and Swachhata-MoHUA already exist; CivicFlow must demonstrate clear local-level value (better UX, integrated issue tracking, local engagement) and be positioned as complementary rather than duplicative.^{[13][12][^2]}

13. Roadmap (High Level)

- **Phase 0 – Discovery & alignment:** Stakeholder workshops with MeitY, selected states/ULBs, and central departments; confirm legal and integration constraints.
- **Phase 1 – MVP build & pilot:** Build core civic module, minimal services hub, and ULB admin console; pilot in 1–2 cities.
- **Phase 2 – Scale-out & integrations:** Expand to more ULBs, add more services and deeper integrations where approved; strengthen analytics and dashboards.
- **Phase 3 – Intelligence & automation:** Introduce AI-assisted classification, chatbots, and smarter routing based on historical patterns, following patterns in emerging AI-based civic complaint systems.^[^15]

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